7
Membranes for Promoting Partial Oxidation Chemistries

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7.1
Introduction

Oxygen transport membranes (OTMs) most frequently consist of dense mixed metal oxide ceramics which facilitate the exclusive mediation of oxygen, as oxygen anions, from the atmosphere to a desired reaction site. As a consequence, many value added partial oxidation chemistries become facilitated while at the same time the need for an oxygen separation plant is eliminated, giving a significant reduction in overall production costs compared to currently available technology. The ability to promote partial oxidation chemistry is, of course, in part dependent upon the residence time of the reactant at the membrane oxidation surface – for most hydrocarbons, too long a residence time leads to complete combustion. In short, this membrane technology integrates atmospheric oxygen separation and chemical processing into a single step, resulting in the opportunity for a simpler and less expensive overall process. An inherent additional feature of chemical processing using such membrane technology is that NO_x emissions are essentially eliminated since oxygen is separated from nitrogen in the air via the dense nonporous membrane. Furthermore, the size of hardware required for chemical processing will be significantly less than for conventional technology as a consequence of nitrogen being absent in the oxidation compartment of the chemical reactor. This provides a number of clear economic incentives to implement this technology commercially in the near-term.

The dense ceramic membranes used are mixed oxygen anion and electron conductors. When operating under dynamic conditions, oxygen anion transport proceeds from the reducing side, where molecular oxygen from the atmosphere undergoes a four-electron reduction reaction to form two oxygen anions. These oxygen anions pass to the membrane oxidizing surface where oxidation chemistry occurs. Electrons simultaneously move in the opposite direction from oxidizing to reducing membrane surfaces. The principles behind this technology for promoting value added partial oxidation chemistry are illustrated schematically in Figure 7.1. The reaction product at the membrane oxidation side is influenced by catalysis, reactant residence time, oxygen flux, reactant concentration, pressure

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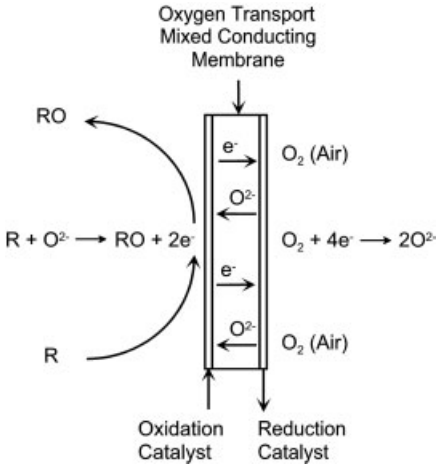


Figure 7.1 Principles behind dense oxygen transport membranes for promoting spontaneous value added partial oxidation chemistry.

and temperature. In particular, OTMs provide a unique opportunity to explore catalysis where oxygen is transported directly to the heterogeneous reaction site as oxygen anions. This allows one to employ catalysts which are lattice matched to the dense membrane and also possess mixed oxygen anion and electron conductivity themselves. This gives a rugged interface between the reactant and membrane while obviating mass transport issues associated with the migration of reactant and molecular oxygen to a heterogeneous catalyst site, as occurs in more conventional chemical processes.

A feature of OTMs that distinguishes them from co-feed catalytic oxidation systems is that, on the fuel side, molecular oxygen in the gas phase is essentially absent. This feature greatly suppresses undesired destructive side reactions in the gas phase, enhancing selectivity. Additionally, the surface coverage of active oxygen species is controlled by the feed partial pressure (hydrocarbon). This tends to suppress deep oxidation and also improve selectivity. Thus, the use of OTMs can result in more selective partial oxidation processes. Also, equilibrium limited reactions, such as methane coupling and alkane dehydrogenation, are spontaneous and complete when carried out as hydrogen abstraction reactions.

The principles behind this membrane technology originate from solid-state electrochemistry. Conventional electrochemical half-cell reactions can be written for chemical processes occurring on each respective membrane surface. Since the general chemistry under discussion here is thermodynamically downhill, one might view these devices as short-circuited solid oxide fuel cells (SOFCs), although the ceramics used for oxygen transport are often quite different. SOFCs most frequently use fluorite-based solid electrolytes – often yttria stabilized zirconia (YSZ) and sometimes ceria. In comparison, dense ceramics for membrane applications most often possess a perovskite-related lattice. The key fundamental

difference between the membranes under discussion here and solid electrolytes applied in electrochemical devices is that, in the former, electron transport proceeds through the membrane lattice bulk, whereas, in a conventional electrochemical process, where the solid electrolyte is an exclusive ionic conductor, electron transport proceeds via an external conduction pathway from the anode, where oxidation occurs, to the cathode where a corresponding reduction reaction proceeds. This raises an important feature of membrane based chemical processes compared to fuel cells. In the former, there is little chemical inhomogeneity across the membrane surface because both oxygen anion and electron transport proceed uniformly through the membrane bulk. In the fuel cell case, electron transport from the anode to cathode requires intimate electronic current collection, ideally from a multiplicity of sites on each respective electrocatalyst (electrode) surface. Since electrochemistry requires electrons, oxygen anions and reactant in close proximity, reaction rates close to a given current collector region are often significantly higher than at sites further away from the point of electron supply or removal. This can lead to significant variations in localized reaction rates from point to point across an electrode surface. Such chemical inhomogeneity across a fuel cell electrode surface can contribute to polarization effects and reduced reaction rates, and lead to material instability issues.

Although there is often a concentration gradient within a membrane reactor, as a given reactant becomes depleted along a reaction channel between reactor inlet and outlet giving rise to a variation in chemical reaction rates, even so localized chemical nonuniformity on the membrane surface, of the type discussed above in the fuel cell case, is generally absent. This is because both the membrane and catalyst deposited on its surface are intrinsically both ionic and electronic conductors, thus eliminating localized regions where chemical inhomogeneity may occur. Furthermore, because of the refractory nature of these membrane materials, they have been shown to be of use for operation under high temperature conditions where a significant pressure differential is maintained between, respectively, partial oxidation and air sides of the membrane reactor. For fuel processing applications, the primary incentives for this can often lie more with process engineering considerations than with membrane chemical kinetics. For example, if natural gas is the feedstock being partially oxidized this resource is frequently retrieved at the wellhead at pressures between 250 and 500 psi (1.7–3.4 MPa). Maintaining that pressure during the spontaneous conversion of natural gas to synthesis gas is important if we subsequently wish to use the partial oxidation product to synthesize liquid fuels via Fischer–Tropsch chemistry where pressure in that general range is desired. Alternatively, if the goal is to produce hydrogen as a feedstock, synthesis gas under pressure, following water-gas shift chemistry, can be introduced into a hydrogen separation membrane where this pressure will not only facilitate hydrogen separation at high rates but will also allow economic geological sequestration of residual concentrated carbon dioxide. Options for operating membranes under differential pressures are summarized schematically in Figure 7.2.

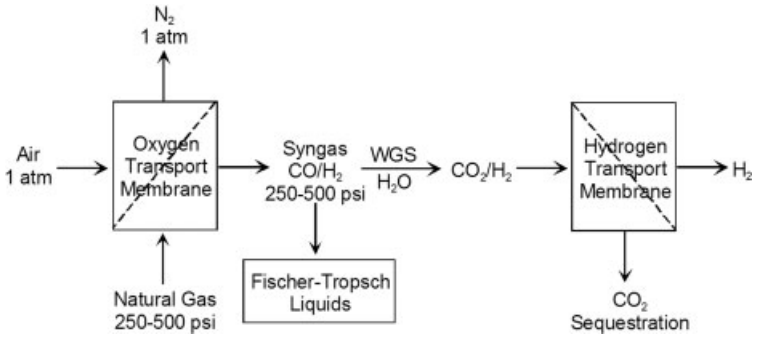


Figure 7.2 Examples of processes benefiting from operating membranes under differential pressure.

Following a brief description of some selected technical criteria which may be used for the rational selection of oxygen mediating membranes, a discussion follows on some partial oxidation chemistries which we hope will illustrate the potential of this OTM technology.

7.2
On the Nature of Perovskite-related Metal Oxides for Achieving Mixed Oxygen Anion and Electron Conduction

7.2.1
Background

In general, the ability to achieve progressively increasing oxygen anion conductivity in metal oxides based upon perovskite-related crystal structures is inversely related to their overall thermodynamic stability. As one might intuitively expect, as the metal oxide lattice becomes less refractory, that is, the average metal–oxygen bond energy decreases, it becomes more vulnerable to both chemical and mechanical instability under the frequently severe conditions to which it is exposed within chemical reactors. For example, during the spontaneous conversion of natural gas (methane) into synthesis gas, the severe reducing conditions present at the partial oxidation surface can lead to loss of membrane lattice oxygen. This in turn can lead to lattice expansion, which can result in membrane mechanical rupture. As a consequence, the membrane lattice substituents must possess chemical stability to both the extreme reducing and oxidizing environments to which the membrane is simultaneously exposed. Hence the rational selection of mixed conducting membrane compositions depends not only upon achieving a sufficiently high oxygen flux (a figure of merit of limited usefulness when comparing different membranes), but also on minimizing lattice oxygen loss and achieving overall chemical stability – membrane materials in future commercial reactors must last for years not weeks!

7.2.2
Early Work towards the Selection of Mixed Conductors

The impetus for the current interest in mixed oxygen anion and electron conducting ceramic membranes was a result of the initial work by Teraoka et al. [1–4] in the mid 1980s on cobalt-containing perovskites possessing the general composition $Ln_{1-x}A_xCo_{1-y}B_yO_{3-\delta}$ (Ln = La, Pr, Nd, Sm, Gd; A = Sr, Ca, Ba; B = Mn, Cr, Fe, Co, Ni, Cu). Based upon oxygen flux data, the ionic conductivity of these materials was considerably higher than that found previously for other materials based, for example, on cermets of YSZ. Trends of higher oxygen anion conductivity and lower material stability in these early materials were manifested by progressively increasing A-site substitution, with the highest oxygen transport rates being reported for $SrCo_{0.8}Fe_{0.2}O_{3-\delta}$. However, application of such materials for combined atmospheric oxygen separation and hydrocarbon partial oxidation chemistry within a single membrane reactor has proven difficult to achieve. This has been a consequence of the high fraction of cobalt in the B-lattice site and the high operating temperature (850–900 °C) at which partial oxidation chemistry proceeds. As those familiar with ceramic processing are aware, the oxides of cobalt possess high volatility and under chemical processing conditions can be readily reduced, leading to decomposition of the lattice [5] and subsequent loss of continuous oxygen transport across the membrane.

To address long-term stability issues with mixed conducting membranes, other early workers investigated a variety of strategies to introduce electronic conductivity into YSZ. This included doping with titania [6, 7], ceria [8], metals such as platinum and palladium [9, 10], and praseodymium-doped indium oxide [11]. The presence of a significant fraction of metallic second phase, to introduce electronic conductivity into an ionically conducting matrix, can, however, lead to long-term mechanical stability issues because of possible thermal expansion mismatch between phases. As a consequence, much of the current activity in this area is focused upon essentially single phase ceramic materials, particularly those possessing perovskite-related crystal structures. When second phases are intentionally introduced into perovskite-based membranes, they are closely matched in expansion characteristics to the bulk material and are present only to enhance membrane ruggedness [12–15].

7.2.3
Requirements for Oxygen Anion and Electronic Conduction within Perovskites

Perovskites possess the general chemical formula ABO_3 and consist of a cubic array of corner-sharing BO_6 octahedra. Oxygen anion transport through a cubic perovskite lattice proceeds via adjacent oxygen edge sites [16]. Perovskite lattice stability requires the existence of a BO_3 skeletal subunit. Geometric considerations dictate that the B cation radii must be greater than 0.051 nm in order to achieve octahedral coordination. Further stabilization of the BO_3 subunit can be achieved by introducing large cations in cubooctahedral coordination at the center

of eight BO_6 octahedra. The BO_3 unit can become distorted by the A-site cation so that optimal A–O bond lengths can be attained. Under conditions of high distortion, orthorhombic or rhombohedral lattice geometries can occur. The radius of cations that can be incorporated into A and B perovskite lattice sites may be defined by the Goldschmidt tolerance factor, S, [17] as represented by:

$$S = -\frac{(r_A + r_o)}{\sqrt{2}(r_B + r_o)} \tag{7.1}$$

where r_A, r_B and r_o correspond to the respective ionic radii of the lattice constituents. From this relationship an ideal cubic perovskite-related structure would possess a tolerance factor of 1.

In practice, however, cubic perovskites are observed possessing tolerance factors between 1.04 and 0.95. Stability for the perovskite structure requires that $1.0<S<0.75$ and that the A lattice site be stable to twelvefold coordination [18]. This requires that the A-site cation possesses a radius greater than 0.09 nm. Tolerance factors between 0.9 and 0.75 promote buckling of the corner-shared octahedra [19] giving rise to many perovskites with an orthorhombic structure.

For oxygen anion transport in mixed conducting membranes, higher oxygen fluxes would in principle be favored by a cubic structure, since in proceeding from cubic to orthorhombic structures the number of crystallographically equivalent sites favoring ionic transport decreases [20]. A further potential feature of the cubic lattice structure for mixed conducting membranes would be the greater facilitation for electronic transport via multivalent transition metals in the B-lattice site compared to the orthorhombic structure. Oxygen anion conductivity, σ, is given by the relationship [21]:

$$\sigma T = A\exp(-\Delta H_m/kT) \tag{7.2}$$

where:

$$A = N\frac{q^2A_o^2}{6k}zfc(1-c)\upsilon_o\exp(\Delta S_m/k) \tag{7.3}$$

and $-\Delta H_m$ (equivalent to E_A) and ΔS_m are, respectively, enthalpies and entropies for activation, N the equivalent sites per unit volume, q the electronic charge, A_o the jump distance, z the number of equivalent near neighbor sites, f a correlation factor (representing deviation from randomness), c the fraction of occupied sites, $1–c$ the fraction of unoccupied sites, υ_0 the lattice vibration frequency, k Boltzmann's constant, and T the absolute temperature. From relationship (7.2), it can be seen that overall ionic conductivity is influenced by the pre-exponential factor, A, which is related to the number of lattice sites available for ionic transport.

One strategy for increasing the population of mobile oxygen anions in a perovskite lattice, so as to potentially increase ionic conductivity, is to enhance the population of lattice oxygen vacancies. This requirement can be realized in the orthorhombic brownmillerite structure [22, 23] which is derived from the perovskite structure by removing one-sixth of the unit cell oxygen atoms to give the empirical

composition $A_2B_2O_5$. The ordered oxygen vacancies in this structure frequently become disordered at high temperature.

The use of an oxygen deficient perovskite lattice has two potential advantages. First, there is an intrinsic population of oxygen vacancies within the lattice bulk as a consequence of the oxygen deficiency. This avoids potential trap sites for ionic migration [24] caused by aliovalent dopants frequently introduced into many ionically conducting materials. Secondly, it is generally more difficult to remove additional lattice oxygen from the membrane under the chemically reducing conditions present at its partial oxidation surface.

7.2.4
Empirical Factors Relating to Oxygen Anion Transport in Perovskite-related Membranes

Until now, it has proven somewhat difficult to develop quantitative predictive tools correlating perovskite-related crystallographic features with their ability to achieve high ionic conductivity. Early work towards this goal addressed modeling ionic conductivity by the application of static lattice simulation techniques [24–26]. Such a strategy assumed a rigid lattice where the sum of all interaction energies between the mobile ionic species and lattice substituents were used to predict the ionic conductivity. Subsequent work refined this approach by incorporating lattice relaxation at the migratory O^{2-} saddle point [27]. Three parameters have been identified which may be empirically correlated to the activation energy for oxygen anion transport in perovskite-related lattices [28]. These are respectively the lattice metal–oxygen bond energy, the degree of openness of the perovskite lattice, or free volume, and the size of the vacancy sites, which must be large enough to permit migration of oxygen anions without significant lattice perturbation.

The average metal–oxygen bond energy (ABE) within a perovskite lattice is estimated by:

$$\mathrm{ABE} = \frac{1}{12\ \mathrm{m}}\left(\Delta H_{A_mO_n} - m\Delta H_A - \frac{n}{2}D_{O_2}\right) + \frac{1}{6\ \mathrm{m}}\left(\Delta H_{B_mO_n} - m\Delta H_B - \frac{n}{2}D_{O_2}\right) \quad (7.4)$$

where $\Delta H_{A_mO_n}$ and $\Delta H_{B_mO_n}$ are the heats of formation of the A_mO_n and B_mO_n oxides [29], respectively, ΔH_A and ΔH_B are the heats of sublimation [30] of the metals A and B, respectively, D_{O_2} is the molecular oxygen dissociation energy, and m is a stoichiometric factor. As one might anticipate, as the ABE for perovskites decreases so does the activation energy for oxygen anion transport. Unfortunately, this also leads to progressively more unstable materials. Because of the frequently severe chemical environments present on each membrane surface in operating devices, a compromise must be made between simultaneously achieving both high oxygen anion mobility and membrane thermodynamic stability.

The degree of lattice openness can also influence the ease with which oxygen anions can migrate through the lattice. This can be reflected by the free volume, FV, calculated from the relationship:

$$\mathrm{FV} = a^3 - \frac{4}{3}\pi \sum_{i=1}^{3} r_i^3 \tag{7.5}$$

where a is the cubic unit cell length, and the index i runs over all atoms in the unit cell, using tabulated ionic radii values, r [31]. For perovskite-related crystal structures, the general trend is that the activation energy for oxygen anion transport decreases with increasing lattice free volume [28].

It has been shown in perovskites that a saddle point exists [16, 32] for oxygen anion migration in a triangle between two A-site cations and one B-site cation. The radius of the circle touching these three cations is defined as r_C. Clearly oxygen anions possessing a radius $<r_C$ will migrate through this saddle point without disturbing these boundary cations. When the mobile oxygen anion is $>r_C$ some distortion of either the mobile anion or lattice must occur to permit ionic transport. Hence saddle point dimensions would be expected to influence ΔE for oxygen anion migration. This expectation is supported where ΔE decreases with progressively increasing r_C [28].

7.2.5
Introducing Electronic Conductivity into a Perovskite-related Lattice

In single phase metal oxide membranes, electronic conductivity is typically introduced into the material via multivalent transition metals on the B-lattice site [18]. For membranes operating under dynamic conditions during partial oxidation chemistry, oxygen anions move from reducing to oxidizing membrane surfaces while electrons simultaneously migrate in a counter direction. Electron transport proceeds through overlapping B–O–B bonds – the Zener double exchange process [33] which may be represented by:

$$\mathrm{B}^{n+}-\mathrm{O}^{2-}-\mathrm{B}^{(n+1)+} \rightarrow \mathrm{B}^{n+}-\mathrm{O}^{-}-\mathrm{B}^{n+} \rightarrow \mathrm{B}^{(n+1)+}-\mathrm{O}^{2-}-\mathrm{B}^{n+} \tag{7.6}$$

For a given transition metal within the perovskite lattice, electron transport is promoted by overlap between the transition metal B-site and O^{2-} orbitals. This is maximized when the B–O–B bonds are linear, corresponding to a cubic lattice. While the orthorhombic lattice has a lower B–O–B bond overlap, resulting in reduced electronic conductivity, many of the candidate oxygen transport membranes being developed do in fact possess this lattice.

Since mixed conducting membranes operate at elevated temperatures, this additionally results in the thermal promotion of electrons from the perovskite valence band (oxygen 2p orbitals) to the conduction band (t_{2g} orbitals of the B-site cation). This electronic conduction has been characterized as p-related [34]. One would anticipate that electronic conductivity would increase with progressively decreasing perovskite band gaps [35].

7.3
The Application of Oxygen Transport Membranes to Partial Oxidation Chemistries

Here will be discussed partial oxidation chemistries that can be spontaneously promoted across OTMs. Probably one of the more important potential applications of this technology is in the conversion of hydrocarbon and carbonaceous materials into synthesis gas – a precursor for liquid fuel synthesis, hydrogen and a feedstock in its own right for electrochemical oxidation in the anode compartment of a SOFC. However, there are a number of other potentially commercially significant partial oxidation applications of this technology, which include alkane oxidative dehydrogenation to olefins and hydrogen sulfide processing. Technical issues and membrane performance are now discussed for each of these applications.

7.3.1
Natural Gas Conversion to Synthesis Gas – General Considerations

One of the most commercially desirable applications for OTMs is the spontaneous partial oxidation of natural gas into synthesis gas. This latter feedstock is of major importance for its subsequent conversion to methanol and to hydrogen. Additionally, synthesis gas can be converted, via Fischer–Tropsch chemistry, into liquid fuels (gas to liquids, GTL) particularly under circumstances where natural gas is located remotely from the place of consumption. Because the resulting liquid fuels possess an intrinsically higher energy density than natural gas, they would be less expensive to transport. One prime example is the Trans-Alaska Pipeline System, which in the future could be used to transport synthetic liquid fuels derived from Alaska's vast reserves of natural gas in the Prudhoe Bay area to the Port of Valdez [36]. This is one of many remote natural gas locations around the world which could take advantage of OTM technology.

A second major driver for the development of low-cost synthesis gas technology from natural gas is the anticipated rapidly emerging hydrogen economy [37] where the hydrogen fraction of the syngas feed can be maximized via water-gas shift chemistry. The resulting hydrogen can then be utilized in fuel cells, direct combustion for energy production, commodity chemicals production such as methanol or ammonia synthesis, or in energy related chemical processing such as petroleum refining, including hydrodesulfurization [38] or hydrocracking [39].

Current commercially available technologies for the production of syngas are either endothermic or require cryogenically separated oxygen – a process which is very energy intensive. They include steam methane reforming (SMR) [40], autothermal reforming (ATR) [41], partial oxidation (POX) [42], and combined reforming (CR) [43]. SMR typically proceeds heterogeneously over nickel on an alumina support via the reaction:

$$CH_4 + H_2O(g) \rightarrow CO + 3H_2; \quad \Delta H = +206 \text{ kJ mol}^{-1} \tag{7.7}$$

which is highly endothermic. If this feedstream is to be used in Fischer–Tropsch (FT) chemistry, the H_2/CO ratio frequently needs to be adjusted to around 2 to promote reaction (7.8) leading to liquid fuels such as gasoline or diesel:

$$2H_2+CO \rightarrow H_2O+1/n(C_nH_{2n}) \tag{7.8}$$

Practically, natural gas following partial oxidation gives essentially the desired H_2 : CO ratio of 2 : 1 and water-gas shift inactive cobalt-based FT catalysts can be used to produce these liquid fuels. However, in the case of coal, water-gas shift is effected *in situ* using an iron-based catalyst. Alternatively, the hydrogen fraction in the syngas feed can be maximized by subsequently applying the exothermic water-gas shift reaction (7.9):

$$CO + H_2O(g) \rightarrow CO_2+H_2; \quad \Delta H = -41 \text{ kJ mol}^{-1} \tag{7.9}$$

The other processes mentioned above (ATR, POX and CR) are all net exothermic reactions but rely upon cryogenically separated oxygen – an expensive and energy intensive process.

In comparison, catalytic membrane reactors (CMRs) based upon OTM technology integrate oxygen separation from the atmosphere and hydrocarbon processing into a single exothermic step (see Fig. 7.1):

$$CH_4 + 1/2\ O_2 \rightarrow 2H_2 + CO \tag{7.10}$$

Technical material issues that are currently being addressed to achieve commercially relevant performance for the OTM-based technology are: (i) the ability to mediate a sufficiently high flux of oxygen across the membrane; (ii) achieving this with membrane materials which possess both high stability and acceptably low coefficients of thermal and chemical expansion; (iii) designing catalysts, particularly on the membrane partial oxidation surface, which exploit the empirical reaction path followed in the conversion of natural gas to syngas; (iv) identifying a reactor design which is compatible with sustaining a high pressure differential between respective hydrocarbon feed and air sides; (v) identifying a seal strategy between respective membrane compartments which enables membrane operation at high differential pressure; and (vi) selecting a reactor design which permits the convenient introduction and removal, respectively, of reactants and reaction products without inducing undo stress on the material components.

An early demonstration of the longevity that can be realized from OTMs was realized using [44] membrane materials possessing the general brownmillerite derived composition [45–49] $Sr_{2-x}La_xFe_{2-y}Ga_yO_{5+\delta}$, where $0<x<1$ and $0<y<1$. Membrane reactors operating under ambient pressure conditions were operated continuously for over 1 year. The reactor performance was stable over the major portion of this operating period. The syngas production rate was 50 mL min^{-1} cm^{-2}, which corresponds to an equivalent oxygen flux of 9.0 mL min^{-1} cm^{-2}. The H_2 : CO ratio was 1.9–2.0 with syngas selectivity and methane conversion being greater

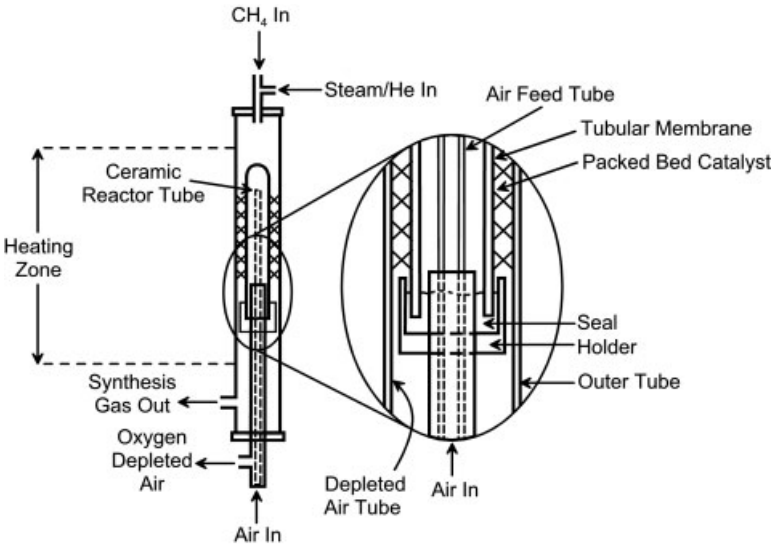


Figure 7.3 Schematic of ambient pressure membrane reactor configuration for promoting the spontaneous conversion of natural gas into synthesis gas.

than 90% and 70% respectively. These ambient pressure reactors utilized closed-one-end tubes prepared using conventional isostatic pressing techniques and were performance tested in a reactor with the configuration shown in Fig. 7.3.

Successful operation of such membrane reactors often requires the hydrocarbon feed to be introduced on the outside wall of the ceramic membrane. This is because the chemically reducing environment on the membrane partial oxidation surface leads to some loss of lattice oxygen, resulting in lattice expansion. Such lattice expansion is more easily managed as a compressive rather than tensile stress, as would occur if the hydrocarbon feed were introduced within the membrane tube. This strategy drastically lowers the incidence of membrane rupture.

Increasing the refractory nature of OTMs generally has the effect of lowering the coefficient of thermal expansion – a desirable feature when one looks toward scaling up this technology to the commercial scale. As discussed earlier, however, this effective increase in the metal–oxygen bond energy results in lowering the intrinsic oxygen anion conductivity, leading to lower oxygen flux rates. This compromise in material properties has resulted in the development [14, 15, 50, 51] of membrane materials possessing the two related compositions $La_{1-x}Ca_xFeO_{3-\delta}$ where, $0<x<0.5$, and $La_{2-a-b}Ce_aCa_bFe_{2-x-y}Mg_xAl_yO_{5+\delta}$, where, $0<a<0.2$, $0<b<0.4$, $0<x<0.3$, $0<y<0.3$. The introduction of cerium into the A-lattice site of the latter material is a strategy for creating intrinsic oxidation catalysis within the membrane.

This type of membrane composition has been used in membrane reactor hardware where a significant differential pressure gradient exists between the hydro-



Figure 7.4 Typical experimental hardware for operating membranes at a high differential pressure. Left: Membrane reactor components. Right: Steam condensation tanks from syngas product. (Photo courtesy of Eltron Research Inc.).

carbon feed (250–400 psi) (1.7–2.6 MPa) and air. In OTM reactors, there is little chemical kinetic advantage to operating at such high pressures. The advantage lies in being able to maintain this pressure from the natural gas wellhead through to either Fischer–Tropsch chemistry or for hydrogen separation. See Fig. 7.4.

7.3.2
Methane Partial Oxidation to Synthesis Gas in Membrane Reactors

In conventional methane heterogeneous partial oxidation chemistry, following mediation of methane and oxygen to a reaction site, reaction leading to synthesis gas has been suggested to proceed via the one-step process [52]:

$$CH_4 + 1/2\ O_2 \rightarrow CO + 2H_2 \tag{7.11}$$

Other work [53] has suggested that reaction proceeds via initial total methane combustion combined with steam reforming and reverse water-gas shift chemistry:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \tag{7.12}$$

$$CH_4 + H_2O \rightarrow CO + 3H_2 \tag{7.13}$$

$$CO_2 + H_2 \rightarrow CO + H_2O \tag{7.14}$$

at equilibrium, the net result gives the same CO/H_2 ratio shown in reaction (7.11).

Alternatively, within membrane reactors, oxygen anions migrate to the oxidation surface where methane reaction to syngas proceeds. Because, as we have discussed previously, interfacial chemistry within these mixed conducting metal oxides is electrochemical in nature, the initial reaction step expected will be that which possesses the most rapid kinetics or, in electrochemical terms, has the lowest overpotential. Because methane is a highly refractory hydrocarbon, direct electrochemical oxidation of this species is difficult. However, because natural gas entering the membrane oxidation compartment will always contain some moisture, this will facilitate some level of methane steam reforming at the typical membrane operating temperature of 900 °C. Hence the next step is probably hydrogen oxidation via:

$$H_2 + O^{2-} \rightarrow H_2O + 2e^- \tag{7.15}$$

which proceeds via the adherent catalyst on the membrane surface. It is probably this rapid reaction, which results in the maintenance of a large oxygen partial pressure gradient across the membrane, that permits oxygen fluxes to proceed at practical rates. Exothermicity from this reaction then drives additional steam methane reforming at separate porous catalytic sites in close proximity to the adherent catalyst site. It is probably the combination of these two overall reaction steps which leads to the methane partial oxidation syngas reaction product. A schematic illustrating membrane catalyst configurational requirements for promoting this reaction is shown in Fig. 7.5.

This general reaction pathway is supported by: (i) hydrogen oxidation proceeding rapidly, even in the absence of intentionally introduced adherent catalysis on the mixed conducting membrane oxidation surface, i.e., no extraordinary catalyst

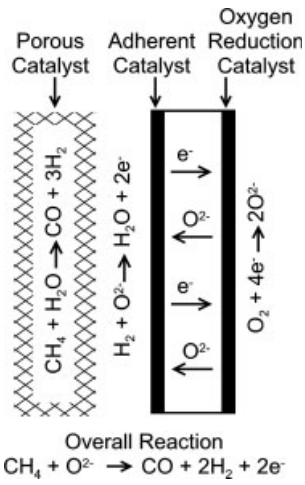


Figure 7.5 Probable reaction sequence for spontaneous natural gas conversion to syngas across an oxygen transport membrane.

requirements are necessary to promote this reaction since the oxidation kinetics are rapid, leading to low surface overpotential losses; (ii) dry methane is relatively unreactive to oxygen anions transported to the membrane oxidation surface – this is consistent with observations found in the anode compartment of SOFCs; (iii) introduction of steam reforming catalysts as a packed bed in close proximity to adherent catalyst on the membrane oxidation surface is observed to increase syngas production rate from natural gas.

7.3.3 Liquid Fuel Reforming

Oxygen transport membranes provide an option for the reforming of liquid hydrocarbon fuels into a carbon monoxide/hydrogen feed [54] which is then compatible for subsequent electrochemical oxidation in the anode compartment of a SOFC. Such a strategy addresses fuel storage issues for fuel cells, particularly in transportation applications.

The reforming of diesel or other hydrocarbon fuels into CO/H_2 mixtures in such an OTM-based reactor may be summarized by:

$$C_nH_{2n+2} + n/2O_2 \rightarrow nCO + (n+1)H_2 \tag{7.16}$$

In order to avoid carbon deposition, steam usually needs to be intentionally co-fed with the hydrocarbon vapor introduced into the reactor. Following initial fuel steam reforming, hydrogen, and to a lesser extent CO, are adsorbed on the high surface area fuel-side catalyst and react with oxygen anions to form H_2O and CO_2. The exothermic oxidation reactions produce heat which drives endothermic steam reforming reactions in a packed bed of catalyst adjacent to the membrane. Steam formed over the oxidation catalysts diffuses to the packed bed of steam reforming catalysts and reacts with hydrocarbon fuel by the reaction:

$$C_xH_y + xH_2O \rightarrow xCO + (x + y/2)H_2 \tag{7.17}$$

The CO_2 reacts with hydrogen:

$$CO_2 + H_2 = CO + H_2O(g) \tag{7.18}$$

Deep oxidation to H_2O and CO_2 is not a concern, because thermodynamics predicts that under equilibrium at high temperatures only H_2 and CO will be stable, as long as the temperature and ratio of C : O : H are properly controlled. The overall process is summarized in Fig. 7.6.

A schematic of the principles behind integrating an OTM-based hydrocarbon fuel reformer with a SOFC for the non-Carnot limited conversion of this feedstock into electricity is shown in Fig. 7.7.

To address sulfur tolerance issues, adherent catalyst on the OTM oxidation surface may consist of cobalt supported on $La_{0.8}Sr_{0.2}CoO_3$, whereas catalysis for the

packed bed region may consist of cobalt impregnated onto granular alumina. At 900 °C, synthesis gas production rates approaching 40 mL min^{-1} cm^{-2} were achieved when converting dodecane as a simulant for logistic fuel. Throughput conversions were 99%. This corresponded to an oxygen flux rate across the membrane >6.3 mL min^{-1} cm^{-2}. Using diesel fuel containing 400 ppm sulfur, the syn-

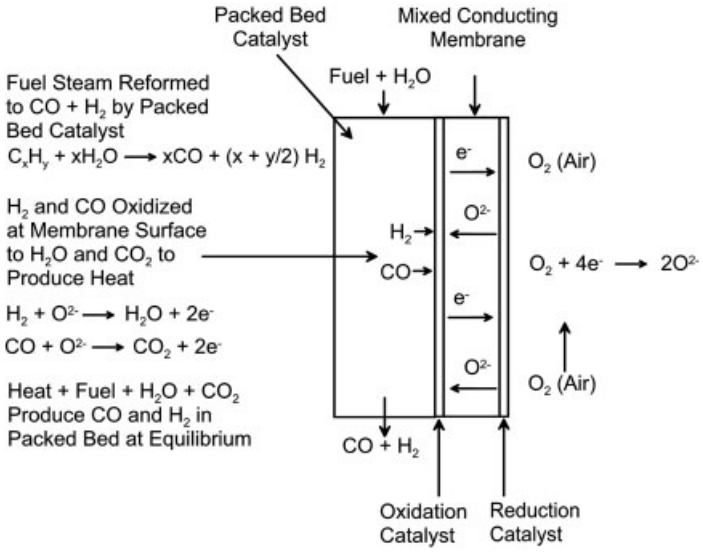


Figure 7.6 Processes occurring during reforming of hydrocarbon fuels to synthesis gas in an OTM reactor.

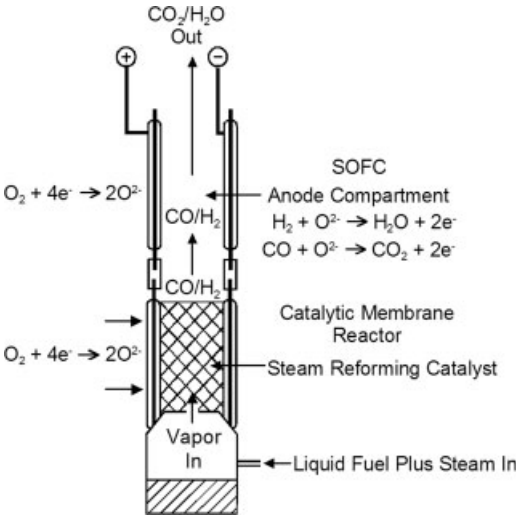


Figure 7.7 Schematic of OTM fuel reformer integrated with a SOFC.

thesis gas production rate was >27 mL min^{-1} cm^{-2} with stoichiometric (100%) throughput conversion. The electrochemically oxidizable fuel in a SOFC would produce an equivalent current of 3.9 A cm^{-2}. Assuming the SOFC operates at 0.75 V per cell, this would correspond to a power output of 2.9 W cm^{-2} of membrane reactor surface area.

7.3.4 Coal/Biomass to Synthesis Gas

The use of adherent refractory mixed conducting catalysts on the partial oxidation surface of OTMs permits gasification of carbonaceous solid feedstocks such as coal fines and biomass to be realized. Refractory catalysts can be selected which not only mediate oxygen to the reaction site from the membrane bulk, but more importantly can be tailored to have high resistance to potentially poisoning impurities, typified by sulfur-containing materials, present in the gasified feedstock.

The spontaneous partial oxidation of coal fines into synthesis gas can proceed [55] by introducing this feedstock with steam into the membrane oxidation compartment as illustrated schematically in Fig. 7.8.

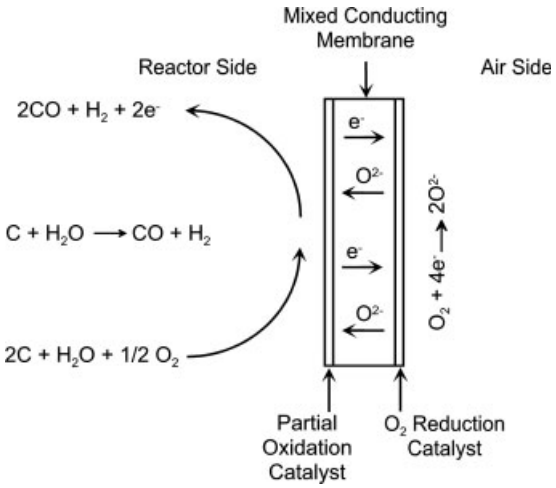


Figure 7.8 Illustration of processes occurring during gasification of coal-steam feedstreams to synthesis gas in a ceramic membrane reactor.

While the exact gasification pathway is not yet clearly understood, it is almost certain that it involves devolatilization, followed by char gasification, and that the latter is initiated by either: (i) initial endothermic coal steam reforming, or (ii) direct exothermic reaction between molecular oxygen mediated across the membrane and coal – possible reactions are as follows:

$$C + H_2O \rightarrow H_2 + CO \tag{7.19}$$

$$C + 1/2O_2 \rightarrow CO \tag{7.20}$$

$$C + O_2 \rightarrow CO_2 \tag{7.21}$$

Subsequent oxidation of hydrogen and carbon monoxide via membrane transported oxygen, leading to more steam and carbon dioxide leads to further coal reforming. For the process to be self-sustaining it needs to be operated at the thermoneutral point or slightly exothermically. Figure 7.9 shows a plot of the enthalpy of the reaction mixture as a function of the steam to carbon ratio, holding the amount of oxygen constant, at a temperature of 1200 K. Variations in temperature by several hundred degrees have very little effect on the trend displayed in this figure. As can be seen, the addition of steam results in the overall reaction becoming less exothermic, until the thermoneutral point occurs at a H_2O/C ratio of about 0.45. This corresponds to a feedstream of C : O_2 : H_2O in the ratio of 1.8 : 0.5 : 0.8 as given by reaction (7.22):

$$1.8C + 0.5O_2 + 0.8H_2O \rightarrow 0.8H_2 + 1.8CO \tag{7.22}$$

which yields a product gas with a H_2:CO ratio of 0.44 : 1.

Operating these reactors at 900 °C, syngas production rates corresponding to up to 20 mL min^{-1} cm^{-2} of membrane surface area were realized over several hundred hours using high sulfur coal (Illinois #6). An interesting consequence of this strategy was that gasification proceeded below slagging temperatures (where coal ash becomes molten) so that coal ash residue could be readily removed from the membrane oxidation compartment. Integration of this membrane-based gasifier with a rehumidifier downstream and water-gas shift catalysis led to near theoretical amounts of hydrogen from coal. Introduction of up to 25 vol% wood fiber or

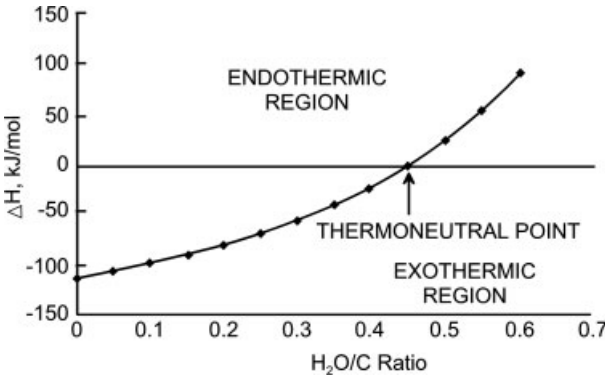


Figure 7.9 Dependence of the enthalpy for simultaneous coal partial oxidation and steam reforming reactions on the steam to carbon ratio at a temperature of 1200 K.

other biomass sources combined with coal as the primary feedstock had little impact on membrane gasifier performance.

7.3.5 Oxygen Reduction Catalysis Requirements in Oxygen Transport Membranes

Oxygen separation from the atmosphere requires membrane catalysis interfacial requirements compatible with promoting initial oxygen dissociative adsorption at the reducing surface and subsequent electron transfer leading to formation of oxygen anions (O^{2-}). This may be represented by Eq. (7.23):

$$O_2 + 4e^{/} + 2V_O^{\bullet\bullet} \rightarrow 2O_O^x \tag{7.23}$$

where $V_{ö}$ represents an oxide ion vacancy in a normal lattice position, where such vacancy sites may be expected to be compatible towards stabilization of O^{2-} species. Requirements for promoting oxygen reduction on the membrane surface may be appreciated by breaking down the overall reduction reaction into the following distinct steps:

1. $O_{2(g)} \rightarrow O_{2ads}$ Initial oxygen adsorption on the catalyst site.
2. $O_{2ads} \rightarrow 2O_{ad}$ Oxygen dissociation.
3. $O_{ad} + 2e^{/} + V_O^{\bullet\bullet} \rightarrow 2O_O^x$ Electron transfer to dissociatively adsorbed oxygen.
4. Subsequent O^{2-} migration from the catalyst into the mixed (O^{2-}/e^{-}) conducting membrane bulk for later partial oxidation chemistry.

Ideally, catalysts for promoting the oxygen reduction reaction should possess both mixed ionic and electronic conduction. This is obviously most conveniently achieved by the catalyst itself having a closely related lattice to the membrane bulk. Under such circumstances, initial oxygen adsorption onto the perovskite catalyst would be expected to proceed on the transition metal B site end-on via lone pair electrons to give a σ-related bond with a metal d_z^2 orbital. Simultaneously, a back donation of t_{2g} electrons occurs from the B site to the oxygen antibonding π^* orbital. This synergistic effect results in strengthening the M–O bond and weakening the O–O bond. This process would be expected to be promoted most effectively for catalysts possessing empty metal e_g orbitals and filled metal t_{2g} orbitals. For M^{2+}, M^{3+}, M^{4+} first row transition metals, this requirement is met for the species Fe^{2+}, Co^{3+} and Ni^{4+} immobilized within perovskite B lattice sites. Subsequent net electron transfer from the metal to the dissociatively adsorbed oxygen atoms would be facilitated by the oxygen atoms being energetically close to the perovskite-related catalyst conduction band edge, under conditions where this band edge is negative relative to that of the dissociatively adsorbed oxygen.

7.3.6 Methane to Ethylene

Oxygen conducting membranes provide one with the opportunity to mediate oxygen as oxygen anions directly to catalyst active sites. If the adherent catalyst on the membrane oxidizing surface possesses mixed oxygen anion and electronic conductivity, one has the opportunity to manipulate lattice substituents to promote a given chemistry. For example, if the metal–oxygen bond energy in the adherent catalyst on the membrane partial oxidation surface is relatively weak, leading to high oxygen anion mobility, formation of synthesis gas from methane is the more likely outcome. Conversely, conditions of low anionic mobility within the catalyst, corresponding to stronger metal–oxygen bond energies, can provide an opportunity for other hydrocarbon partial oxidation chemistries to proceed. Consequently, rates of these processes are anticipated to follow the Sabatier principle (i.e., optimal bond strengths). The relative values of M–O bond strengths and, hence, the relative rates of processes determine the overall selectivity. One reaction of particular commercial significance is the oxidative dimerization of methane to ethylene and ethane given by:

$$2CH_4 + O_2 \rightarrow C_2H_4 + 2H_2O(g); \quad \Delta G = -287.6 \text{ kJ mol}^{-1} \tag{7.24}$$

and:

$$4CH_4 + O_2 \rightarrow 2C_2H_6 + 2H_2O(g); \quad \Delta G = -319.9 \text{ kJ mol}^{-1} \tag{7.25}$$

Conventional heterogeneous catalysis approaches to promote this methane coupling chemistry rely upon both methane and oxygen being present as a gas phase co-feed. Using such an approach, both of these species must, prior to reaction, simultaneously diffuse within the gas phase to a catalyst site. The technical issue here is the fact that the rate of reaction product, ethylene, oxidation is greater than that for the more refractory hydrocarbon, methane. Hence, this leads to a limited yield of the desired product (ethylene). In comparison, the promise of membrane-based reactors for promoting this chemistry, is that oxygen becomes selectively mediated to the membrane oxidation catalyst site, which ideally is also a mixed conductor, where the subject methane coupling chemistry proceeds. Using this strategy, the presence of molecular oxygen in the presence of the olefinic reaction product is avoided, thereby giving the opportunity for higher selectivity to ethylene at a given level of methane conversion efficiency.

For the spontaneous conversion of methane to ethylene, for example, the respective empirical oxidation and reduction half-reactions occurring on each membrane surface in these O^{2-} conducting materials may be represented as follows:

At the oxidizing membrane surface

$$2CH_4 + 2O^{2-} \rightarrow C_2H_4 + 2H_2O + 4e^- \rightarrow \text{through membrane to oxygen reduction site} \tag{7.26}$$

At the reducing membrane surface

$$O_2 + 4e^- \rightarrow 2O^{-2} \rightarrow \text{through membrane to methane oxidative dimerization site} \quad (7.27)$$

7.3.7
Catalysis Considerations for Promoting Methane Coupling Reactions

The initial work which promoted interest in conventional heterogeneous catalysis for methane coupling chemistry was first reported in 1982 [56]. In this work, the subject reaction was evaluated using 23 metal oxides supported on α-alumina over the temperature range 500–1000 °C. Methane and oxygen were, however, sequentially introduced into the reaction vessel, suggesting that the overall chemical reaction involved the general Mars–van Krevelen reaction sequence:

$$CH_4 + \text{metal oxide} \rightarrow {}^1/_2C_2H_4 + H_2O + \text{reduced metal oxide} \quad (7.28)$$

$$\text{reduced metal oxide} + O_2 \rightarrow \text{metal oxide} \quad (7.29)$$

Later it was shown that sequential introduction of CH_4 and O_2 was not necessary to promote methane oxidative dimerization but this could be achieved directly by passing CH_4/O_2 mixtures over a metal oxide catalyst [57]. Since these early reports, work directed towards investigating the chemical oxidative dimerization of methane has increased with a significant number of papers [58–88] and reviews [89, 90] being published.

Catalysis requirements for promoting methane coupling on the oxidizing surface of oxygen anion conducting membranes ideally requires that the reaction site is either a super-base or an amphoteric material, possessing electrophilic (Lewis acidic) as well as basic functions. Methane activation at basic catalyst sites on the membrane surface relies upon O^- species being continuously generated at the catalyst/membrane interface, originating from mobile O^{2-} species mediating through the membrane to the methane partial oxidation interfacial region. Hydrogen abstraction from methane in natural gas to give methyl radicals is thought to proceed via the general reaction [91–107]:

$$M^+O^- + CH_4 \rightarrow M^+OH^- + CH_3\cdot \quad (7.30)$$

where M^+ corresponds to an immobilized cation lattice site within the oxygen anion and electron conducting catalyst on the membrane oxidizing surface. Methyl radicals formed in this manner could then couple in the gas phase [102, 103] via:

$$2CH_3\cdot \rightarrow C_2H_6 \quad (7.31)$$

to give ethane.

The presence of surface O^- species on catalyst sites has been directly observed using electron paramagnetic resonance (EPR) studies on various metal oxides [108] in heterogeneous reactions using CH_4/O_2 mixtures, and chemically shown to be active by reaction of methane with metal peroxides [106] which are the dimeric analogs of O^-.

In contrast to the above described behavior of the basic component of an amphoteric metal oxide catalyst, the electrophilic function interacts with methane in a different manner. Here, the relatively electron rich C–H hydrogen bonds serve as electron donors to the metal center, resulting in a weakening of these bonds and methane activation, leading to formation of ethane or even direct formation of ethylene.

Ethylene can be formed from ethane via a subsequent proton abstraction step involving both mobile and immobile oxygen anion sites at the membrane oxidizing surface, as shown by the reaction sequence below:

$$C_2H_6 + O^- \rightarrow C_2H_5\cdot + OH^- \tag{7.32}$$

$$O^{2-} \rightarrow O^- + e^- \tag{7.33}$$

$$O^- + C_2H_5\cdot \rightarrow C_2H_4 + OH^- \tag{7.34}$$

Ethane contained in natural gas may also contribute to methane coupling chemistry. In the temperature regime of interest in methane oxidative dimerization (~750 °C), ethane undergoes thermal cracking that increases with temperature. This generates an additional supply of methyl radicals via:

$$C_2H_6 \rightarrow 2CH_3\cdot \tag{7.35}$$

This supply of methyl radicals can either recombine to regenerate ethane or abstract hydrogen from ethane to produce ethylene:

$$2CH_3\cdot \rightarrow C_2H_6 \tag{7.36}$$

$$CH_3\cdot + C_2H_6 \rightarrow C_2H_5\cdot + CH_4 \tag{7.37}$$

$$CH_3\cdot + C_2H_5 \rightarrow C_2H_4 + CH_4 \tag{7.38}$$

In addition to the above pathways, which proceed through methyl radicals, oxidative dehydrogenation of ethane in the feedstream, as represented by Eq. (7.32) to (7.34), can also occur.

7.3.8
Catalyst Implementation on Dense Oxygen Transport Media for Oxidative Coupling

Using electrolytic cells based upon YSZ, mixed conducting anode electrocatalysts which showed [109] activity toward promoting the methane coupling reaction included $Sm_{0.5}Ce_{0.5}CuO_3$, $Tb_{0.8}Sm_{0.2}CuO_3$, $Gd_{0.9}Th_{0.1}CuO_3$, $Gd_{0.9}Na_{0.1}MnO_3$, and $Th_{0.8}Yb_{0.2}NiO_3$. Maximum partial faradaic current densities at active perovskite anode electrocatalysts for promoting the subject reaction were found directly correlatable to their calculated oxygen binding energies on the perovskite surface, where increasing binding energies were found to favor higher rates for electrochemical methane coupling. Increasing surface oxygen binding energies at perovskite anode electrocatalysts were found to correlate with increasing perovskite free volumes, supporting increasing surface oxygen binding energies and perovskite free volumes as leading to enhanced rates for the subject reaction. As a consequence, synergism was found between experimentally determined perovskite anode electrocatalyst activities, their calculated surface oxygen binding energies, and free volumes.

Other workers [110] have investigated the influence of mediated O^{2-} species through the oxygen conducting solid electrolyte YSZ for promoting the subject reaction. Using an electrolytic cell of configuration Ag,MgO/YSZ/Ag, passage of CH_4 and O_2 mixtures through the Ag,MgO anode compartment under open-circuit conditions at 700 °C gave trace C_2H_4 and C_2H_6. Upon electrochemically driving O^{2-} through the cell to the Ag,MgO anode, the C_2H_4 yield was found to decrease dramatically with C_2H_6 being unaffected. In comparison, other workers [111] using the cell Ag,Bi_2O_3/YSZ/Ag showed that methane oxidative dimerization, mainly to C_2H_6, could be promoted when only faradaically driven oxygen was available at the Ag,Bi_2O_3 anode.

Methane coupling chemistry has been reported [112, 113] in mixed conducting membranes of $Bi_{1.5}Y_{0.3}Sm_{0.2}O_3$. Coupling selectivities as high as 54% and yields of 35% were obtained at 900 °C. The simultaneous oxidative coupling of methane and oxidative dehydrogenation of ethane over a basic catalyst system has been reported [114], as well as the use of carbon dioxide as an oxidant in the methane/ethane system[115]. In the latter case, ethylene content of up to 16% was obtained in the tail gas.

Using membrane reactors, it has been found that incorporation of 10% ethane into a methane feed on the membrane oxidation surface can lead to a significant increase in ethylene yield [116]. In this work it was found that overall selectivity to ethylene was 94% and methane conversion 36%, indicating that the heuristic conversion + selectivity ≈100 % was significantly exceeded.

7.3.9
Alkane Dehydrogenation

As with the previously discussed methane coupling chemistry, the promise of OTMs for promoting alkane oxidative dehydrogenation reactions such as:

$$C_2H_6 + O_2 \rightarrow C_2H_4 + H_2O \quad (7.39)$$

lies in their potential for overcoming the equilibrium limitation of the conventional dehydrogenation reaction and for achieving higher selectivities and conversion efficiencies of co-feed oxidative dehydrogenation when compared to conventional catalytic reactors. This general expectation has been confirmed by two groups using OTMs based upon the perovskitic membrane $Ba_{0.5}Sr_{0.5}Co_{0.8}Fe_{0.2}O_{3-\delta}$ [117, 118]. The intrinsic catalytic behavior of this membrane [117] gave 80% selectivity for converting ethane to ethylene under conditions where 84% of the ethane was consumed at 800 °C. This compared to 53.7% ethylene selectivity when using a more conventional fixed bed reactor with catalyst material of the same composition. Similar results were obtained by other workers [118] where respective ethylene yields and selectivities of 68% and 92% were obtained from the intrinsic catalysis of this membrane material when operated at a similar temperature. Introduction of dispersed palladium catalysis onto the membrane dehydrogenation surface resulted in ethylene yields and selectivities of 76% and 86% respectively – a significant improvement. However, substitution of palladium by nickel as the catalyst resulted in a decrease in membrane performance for promoting this reaction.

Somewhat similar results have been found [119] for mixed conductors based upon the fluorite structured membrane $Bi_{1.5}Y_{0.3}Sm_{0.2}O_3$ where per pass ethylene yields and selectivities from ethane of 56% and 80% respectively were obtained at 875 °C. Using the same materials in a fixed bed reactor gave ethylene yields and selectivities of 11%.

All of this preliminary reported work on OTMs for promoting ethane dehydrogenation to ethylene supports the merits of this strategy compared to conventional dehydrogenation and oxidative dehydrogenation technologies.

7.3.10
Hydrogen Sulfide Partial Oxidation

Oxygen transport membranes can be used for the single step conversion of a hydrogen sulfide feedstream into elemental sulfur and steam as shown schematically in Figure 7.10.

Here OTMs may present the opportunity to develop a single stage Claus process. In the well established Claus process [120], hydrogen sulfide is initially burned in air to give SO_2 via:

$$2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O \quad (7.40)$$

followed by:

$$2H_2S + SO_2 \rightarrow 3S + 2H_2O \quad (7.41)$$

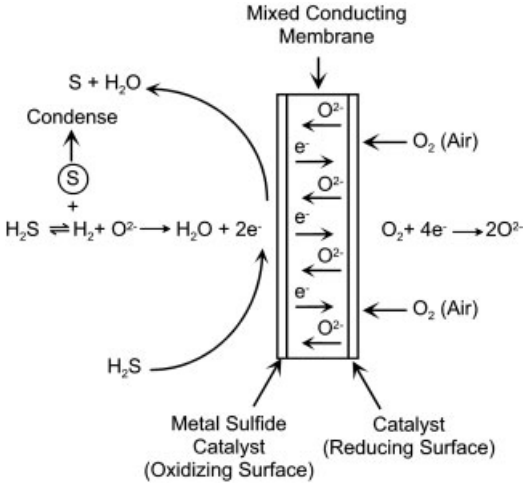


Figure 7.10 OTM for the spontaneous partial oxidation of H_2S to S and H_2O.

with many side reactions being possible. Hydrogen sulfide will dissociate at high temperatures via the equilibrium [121]:

$$xH_2S = xH_2 + S_x \tag{7.42}$$

Additionally, it has been shown that both molybdenum and tungsten disulfides can act as catalytic sites [122] for maintaining this equilibrium. Hence if hydrogen could be selectively oxidized to steam in an OTM, equilibrium (7.42) could be continuously shifted to the right, resulting in the same reaction product in a single step that the Claus process gives in two steps.

The first exploitation of this strategy was performed in SOFCs based upon both YSZ [123–125] and ceria [126]. For one of these fuel cell strategies [125] where metal sulfides were used as anode electrocatalysts, rapid electrode kinetics were found for the hydrogen oxidation reaction with the empirical trend for exchange currents (i_o) per geometric area being found to be: $NiFe_2S_4$>WS_2>$CuCo_2S_4$> $CuFe_2S_4$ = $NiCo_2S_4$>$CuNi_2S_4$.

Using insights derived from the above electrochemistry, OTMs have been operated [15] possessing the general composition $Sr_{2-x-y}La_xCa_yB_zB'_{2-z}O_{5+a}$ where B=Al or Ga and B′ = Mn, Fe, or Co, $0 \le x \le 0.5$, $0 \le y \le 2$, and $0 \le z \le 0.6$. The membrane reactor used the above catalysts adherent to the membrane oxidation surface and a variety of packed bed catalysts located within the reactor oxidation compartment which included 5% CoS_2 on Al_2O_3, and 8% MoS_2 on Al_2O_3.

Hydrogen sulfide conversions of up to 62 mL min^{-1} were obtained at 900 °C with a 40% H_2S stream flowing at 200 mL min^{-1}. This corresponded to an equivalent current density of 710 mA cm^{-2} and close to 550% removal beyond that expected from thermodynamic hydrogen sulfide thermal decomposition at this

temperature. The oxygen flux measured across the membrane from oxygen depletion on the air side corresponded to 4.5 mL min^{-1} cm^{-2}.

An interesting observation was that the presence of metal surface catalysts for the hydrogen oxidation reaction appeared to have little effect upon performance. This was probably a consequence of the hydrogen oxidation electrochemistry occurring at a significantly more positive potential than in the fuel cell case. The supported metal sulfide packing in the oxidation compartment was necessary for rapidly shifting equilibrium (7.42) upon hydrogen depletion. Typical H_2S conversion rates were 95%.

7.3.11
Some Thoughts on the Potential Contribution of Membrane Technology towards Realizing a Hydrogen Economy

Both electrochemical and photoelectrochemical strategies have been extensively investigated as potential environmentally benign approaches for hydrogen production. The attractiveness of electrolytic technology is that it gives one the option to use electrical energy to split water directly into its separate constituents, hydrogen and oxygen. While this is a simple and elegant approach for hydrogen generation, the electrical energy required to drive this process is almost invariably produced by the initial combustion of fossil fuels. The intrinsic inefficiencies associated with the conversion of fossil fuels to electrical energy followed by those of the electrolytic water splitting step in general lead to hydrogen costs approximately four times higher than for natural gas. The use of electrolytic technology to achieve low-cost hydrogen production must probably await low-cost nuclear energy as the electricity source – a capability which only appears somewhat in the distant future at this time.

Photoelectrochemical (PEC) water splitting is, in principle, a particularly attractive approach for hydrogen generation from water. This approach depends upon photons of incident sunlight exciting electrons within a semiconductor particle which itself is in direct contact with water. Provided that the photons possess the appropriate energy, hydrogen and oxygen can be simultaneously liberated from, respectively, the semiconductor conduction and valence bands. In several ways, the overall process is an inorganic analog of photosynthesis, with both oxygen and hydrogen being produced. In photosynthesis, hydrogen-related species participate in reducing carbon dioxide to form carbohydrates, whereas in the PEC case hydrogen is directly liberated. Major issues which limit the commercial practicality of this strategy any time soon are: (i) an inadequate match between semiconductor photon absorption and the predominant wavelengths of incident sunlight, (ii) semiconductor instability, (iii) low power density of incident sunlight which would dictate very high surface areas to achieve hydrogen generation in commercial quantities, and (iv) PEC water splitting would produce hydrogen/oxygen mixtures which introduces both safety and separation issues.

The implication that one might derive from the above is that in the foreseeable future, hydrogen for the Hydrogen Economy will need to rely upon hydrocarbon feedstocks, notably originating from natural gas and coal. Such an approach to the Hydrogen Economy will however produce a significant amount of carbon dioxide. Hence, any technology for hydrocarbon or coal reforming to hydrogen must be compatible for conveniently facilitating carbon dioxide sequestration.

In principle, by combining both oxygen and hydrogen transport membranes, discussed by the many contributors in this book, one might achieve this goal in a spontaneous overall process. This would involve initial reforming of the hydrocarbon or carbonaceous feedstock, using OTM technology, into a mixture of carbon monoxide and hydrogen under pressure. Following water-gas shift, the predominantly carbon dioxide/hydrogen mixture could be introduced into hydrogen transport membranes under pressure, thereby facilitating an opportunity for simultaneous hydrogen separation and carbon dioxide sequestration. The overall process leading to a hydrogen feedstock from such feedstocks is thermodynamically downhill. How hydrogen combustion might be incorporated into an environmentally benign electric utility plant whereby separated carbon dioxide could be used to either cogenerate hydrocarbon feedstock or be sequestered is illustrated in Fig. 7.11.

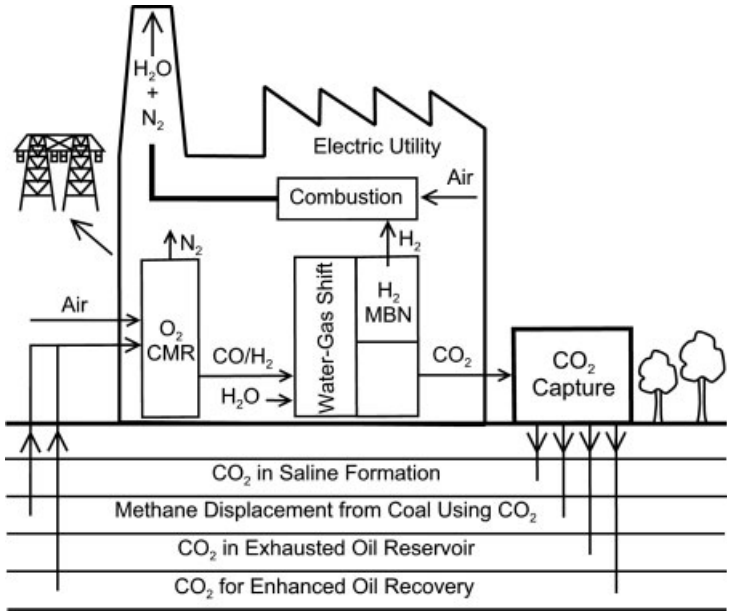


Figure 7.11 Schematic of how OTM based catalytic membrane reactors (CMRs) and hydrogen transport membranes (H_2 MBN) might facilitate an environmentally benign electric utility.

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